FinDoc-AuditEngine v3.0 — Financial Reconciliation & Explainable ML Prototype

Linkific AI/ML Engineering Internship Capstone • Shri Sanjaykumar V (M.Tech Integrated SWE, VIT Vellore)

Working Prototype & Reference Implementation

- Inspired by practical procurement experience managing INR 60+ Lakhs in sponsorships & purchase orders for VIT Riviera & Gravitas.
- Combines Stage 1 Regex PII Masking (6 patterns), Stage 2 Deterministic 3-Way SKU Matching (Invoice vs PO vs GRN), Stage 3 7-Feature IsolationForest anomaly scoring (seeded synthetic baseline N=1,000), and Stage 4 5-Tier Governance Routing.
- Includes local single-feature perturbation attribution (+pts per risk driver) and dual-currency USD/INR display (1 USD = 86.50 INR).

Presenter & Verification Profile

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Candidate: Shri Sanjaykumar V
Institution: VIT Vellore (M.Tech Int. SWE)
CGPA: 9.12 / 10.0
Internship: Linkific (AI/ML Intern)
Day 29 PyTests: 28 passed (all collected)
Day 30 PyTests: 33 passed (all collected)
Ledger: In-memory prototype + CSV export
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STAGE 1 PRIVACY GATE

6 Regex Patterns

PAN, Aadhaar, Account, Card, Phone, Email

STAGE 2 MATCHING ENGINE

3-Way SKU Match

Invoice vs. PO Contract vs. Dock GRN

STAGE 3 ANOMALY MODEL

7-D Isolation Forest

1,000-Record Seeded Synthetic Baseline

AUTOMATED VERIFICATION

28 / 33 PyTests

All Collected Tests Passed via pytest -q

Speaker Cue: Welcome mentors and panel members. Today I will present FinDoc-AuditEngine v3.0, our working prototype for Accounts Payable auditing.

Why Combine Deterministic 3-Way Rules with Unsupervised Anomaly Scoring?

Comparing Manual Review, Static ERP Rules, Standalone LLMs, and Our Hybrid Prototype

Dimension	Manual AP Review	Static ERP Rules	Standalone LLM	FinDoc-AuditEngine v3.0
SKU Price & GRN Math	Slow, manual fatigue	Exact configured rules ($0.01)	Floating-point rounding risk	Deterministic SKU & GRN math ($0.05 tol)
PO-Matched Anomalies	Ad-hoc intuition	Misses offshore rush PO matches	Uncalibrated text opinions	7-D IsolationForest (synthetic baseline)
Risk Attribution (XAI)	Manual notes	Binary pass/fail code	Generated text rationale	Single-feature perturbation (+pts)
PII Protection	Raw PII on screen	Logs raw fields	External API exposure risk	Stage 1 regex masking ([PAN_01])
Ledger & Export	Spreadsheets	ERP database	Chat transcripts	In-memory prototype list + CSV export

Core Architectural Takeaway

• Deterministic 3-way matching and unsupervised anomaly detection solve distinct problems: Stage 2 enforces exact contract rates and dock quantities, while Stage 3 surfaces unusual transactions (like a $49,500 Net-1 offshore invoice) that match their PO on paper.

Speaker Cue: Deterministic matching enforces exact price/quantity tolerances, while our synthetic-baseline Isolation Forest highlights unusual multi-feature combinations.

End-to-End 4-Stage Prototype Pipeline Architecture

Sequential Request Flow from FastAPI Ingestion to Governance Routing & In-Memory Audit Ledger

STAGE 1: PRIVACY SCRUBBER

app/core/privacy.py

6 Regex PII Rules -> [SURROGATE]

STAGE 2: 3-WAY MATCHER

app/services/matcher.py

SKU Price ($0.01) & GRN Net Qty

STAGE 3: ANOMALY + XAI

anomaly_detector.py

7-D IsolationForest + Perturbation

STAGE 4: GOVERNANCE ROUTER

app/services/router.py

5 Tiers + In-Memory AUDIT_LEDGER

Module Path	Core Class / Method	Input / Output Contract	Prototype Implementation Detail
app/models.py	InvoicePayload, PurchaseOrderRecord	Pydantic v2 @model_validator	Verifies subtotal * (1 + tax_rate) == total_amount_usd within $0.05
app/core/privacy.py	FinancialPrivacyScrubber	Raw text -> Masked text + count	Regex masking for PAN, Aadhaar format, Account, Card, Phone, Email
app/services/matcher.py	ReconciliationMatcher.reconcile()	Invoice + PO -> ReconciliationResult	Enforces configured SKU unit rate and dock net quantity rules
app/services/anomaly_detector.py	AnomalyScorer.assess_risk()	7-D vector -> AnomalyAssessment	200-tree IsolationForest fit on 1,000 seeded synthetic records
app/services/router.py	ApprovalRouter & AUDIT_LEDGER	Match + Risk -> GovernanceDecision	Routes to 5 tiers; appends to in-memory Python list (prototype)

Speaker Cue: Every audit request flows through four modular Python services instrumented with stage-level timing telemetry.

Stage 1 Regex Privacy Masking & Stage 2 Deterministic 3-Way Matching

Pre-Inference PII Surrogate Tokenization and Exact SKU-Level Price & Quantity Verification

PII Pattern Category	Configured Regex Structure	Surrogate Token
1. Indian Tax PAN	\b[A-Z]{5}[0-9]{4}[A-Z]{1}\b	[PAN_01]
2. Aadhaar Format	\b\d{4}[\s-]?\d{4}[\s-]?\d{4}\b	[AADHAAR_01]
3. Bank Account / IBAN	(?:ACCT|IBAN|A/C)[:\s-]*[A-Z0-9]{8,22}	[BANK_ACCT_01]
4. Credit Card	\b(?:\d{4}[\s-]?){3}\d{4}\b	[CARD_01]
5. Phone Number	(?:\+?\d{1,3}[\s-]?)?\(?\d{3,4}\)?...	[PHONE_01]
6. Email Address	[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+	[EMAIL_01]

Stage 2: Deterministic 3-Way Reconciliation Math

- Contractual Unit Rate Check: Flags RATE_VARIANCE whenever $P_{inv,i} - P_{po,i} > \$0.01$ per SKU.
- Dock GRN Net Quantity Check: Computes $Q_{net,i} = Q_{received,i} - Q_{rejected,i}$ and flags QTY_SHORTAGE if billed $Q_{inv,i} > Q_{net,i}$.
- Uncontracted SKU Check: Flags any invoice SKU absent from the Purchase Order.

Regex Invoice-Text Parser (/api/v1/audit/parse-raw)

- Extracts Invoice ID, PO Number, Vendor, Country, and USD total from plain-text memos via regex while scrubbing PII.
- Honest Scope Note: Performs regex text parsing on raw text strings (not optical PDF/image OCR).

Speaker Cue: Stage 1 masks six configured regex PII patterns, while Stage 2 verifies every SKU against its PO contract rate and GRN dock log.

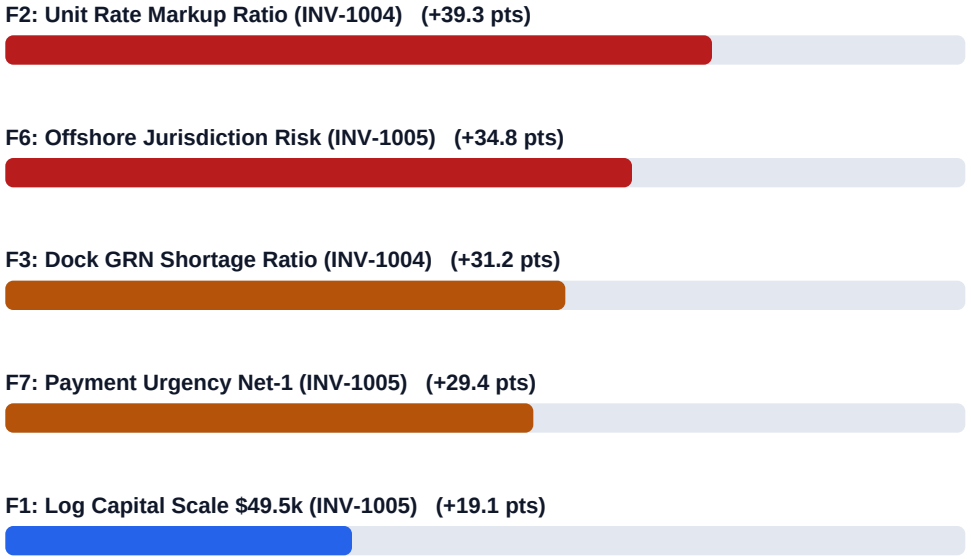
7-Feature Isolation Forest & Local Perturbation Attribution (XAI)

Reproducible Seeded Synthetic Baseline (N = 1,000, seed = 42) & Marginal Risk Point Decomposition

Feature Name	Formula / Transform	Risk Signal Captured
F1: log_total_usd	ln(1 + Total_USD)	Compresses heavy-tailed invoice totals
F2: price_variance_ratio	max(0, (P_inv - P_po)/P_po)	Relative unit rate markup above PO
F3: qty_shortage_ratio	max(0, (Q_inv - Q_net)/Q_inv)	Billing for dock-rejected units
F4: tax_rate_deviation	|tax_rate - 0.18|	Departure from 18% GST baseline
F5: line_item_concentration	1.0 / N_items	Single-line lump-sum concentration
F6: jurisdiction_risk_weight	0.05 (Low) -> 0.85 (High)	Offshore tax-haven jurisdiction risk
F7: payment_urgency_index	1.0 / max(1, terms_days)	Net-1 / Net-2 rush wire urgency

Single-Feature Perturbation Attribution (XAI)

Formula: Delta R_k(x) = max(0, R(x) - R(x^(k <- mu_k)))



Methodological Note: The 0–100 score is an engineered anomaly-risk indicator fit on a 1,000-record seeded synthetic baseline (not a calibrated fraud probability). Production deployment requires validation on real historical company invoices.

Speaker Cue: Our 200-tree IsolationForest is trained on 1,000 seeded synthetic records and paired with local single-feature perturbation attribution.

Scenario 1: Clean Low-Value Procurement Invoice (INV-1001)

Apex Industrial Supplies Pvt Ltd • PO-2026-001 • $432.00 USD (INR 37,368.00) • Tier 1 Auto-Approved

INVOICE & PO REF

INV-1001 / PO-001

Apex Industrial Supplies (IN)

TOTAL (USD / INR)

$432.00 USD

INR 37,368.00 (@ 86.50)

RISK INDICATOR

18.2 / 100 (LOW)

Within synthetic normal baseline

GOVERNANCE ROUTE

TIER 1 AUTO-STP

Authorized = True ($0 Variance)

SKU	Description	Inv Rate	PO Rate	GRN Net
SKU-101	Precision Bearing Assembly	$120.00	$120.00	2 (0 Rej)
SKU-102	Synthetic Hydraulic Fluid 5L	$60.00	$60.00	2 (0 Rej)
Total	Subtotal $360 + 20% Tax	$432.00	$432.00	MATCHED

Terminal Output: Scenario 1 (INV-1001)

```
[STAGE 1] Privacy Scrubber: Masked [EMAIL_01], [PAN_01]
[STAGE 2] 3-Way Matcher: is_matched=True | var=$0.00
[STAGE 3] IsolationForest: risk_indicator=18.2/100 (LOW)
[STAGE 4] Governance Route: TIER_1_AUTO_APPROVED
Authorized: True (Straight-Through)
Dual Currency: $432.00 USD | INR 37,368.00
```

Speaker Cue: Scenario 1 demonstrates a clean invoice ($432.00) where all SKU prices and GRN quantities match PO-2026-001, routing to Tier 1 Auto-Approval.

Scenario 2: Unit-Price Overbilling & Dock GRN Shortage (INV-1004)

Precision Tooling Works • PO-2026-004 • $8,614.00 USD (INR 7,45,111.00) • Rejected Discrepancy

INVOICE & PO REF

INV-1004 / PO-004

Precision Tooling Works (IN)

BILLED VS CONTRACT

$8,614 vs $7,552

+$1,062.00 Total Discrepancy

RISK INDICATOR

79.5 / 100 (HIGH)

Markup (+39.3) + Shortage (+31.2)

GOVERNANCE ROUTE

REJECTED_DISCREPANCY

Authorized = False (Blocked)

SKU	Invoice Billed	PO Contract	GRN Dock Log
SKU-401 Spindle	10 @ $550 ($5,500)	10 @ $450 ($4,500)	10 Rec, 2 Rej (Net 8)
SKU-402 Servo	5 @ $360 ($1,800)	5 @ $380 ($1,900)	5 Rec, 0 Rej (Net 5)
Total (18% GST)	$8,614.00 USD	$7,552.00 USD	REJECTED (+$1,900 base)

Terminal Output: Scenario 2 (INV-1004)

```
[STAGE 2] 3-Way Matcher: is_matched=False (2 Breaches)
- SKU-401 Rate Overbill: Inv $550 vs PO $450 (+22.2%)
- SKU-401 Dock Shortage: Billed 10 vs Net Accepted 8
[STAGE 3] Perturbation XAI: risk_indicator=79.5/100 (HIGH)
- Unit Rate Markup: +39.3 pts
- Dock GRN Shortage: +31.2 pts
[STAGE 4] Route Decision: REJECTED_DISCREPANCY
```

Speaker Cue: Scenario 2 shows deterministic 3-way matching catching both a $100/unit rate markup and 2 dock-rejected units on SKU-401.

Scenario 3: PO-Matched Critical Anomaly (INV-1005 — Frozen Fraud Risk)

Cayman Horizon Advisory Ltd (KY) • $49,500.00 USD (INR 42,81,750.00) • Net-1 Rush Wire

VENDOR & COUNTRY

Cayman Horizon (KY)

High-Risk Offshore Jurisdiction

3-WAY MATCH STATUS

PASSED ($0.00 Var)

1 unit @ $49,500 matches PO-005

RISK INDICATOR

99.0 / 100 (CRITICAL)

Isolated at shallow tree depth

GOVERNANCE ROUTE

FROZEN_FRAUD_RISK

Authorized = False (CFO Hold)

Why Static Rules Pass INV-1005 vs. Why Stage 3 Flags It

• Deterministic Check: SKU-501 bills 1 unit @ $49,500.00 against PO-2026-005 ($49,500.00) with 1 accepted GRN unit -> $0.00 variance.

• Multi-Feature Anomaly Pattern Flagged by 7-D IsolationForest:

• 1. Offshore Jurisdiction Risk (KY): High-risk tax haven (0.85).

• 2. Payment Window Urgency: Demands Net-1 immediate wire.

• 3. Structuring Below Threshold: $49,500 sits just below $50k cap.

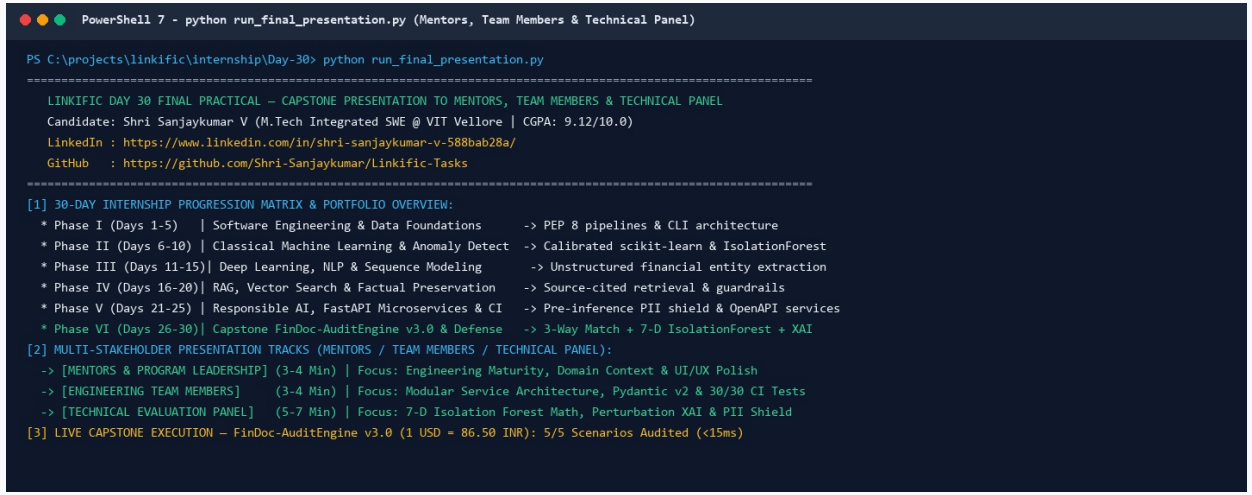
Terminal Output: Scenario 3 (INV-1005)

```
[STAGE 1] Privacy Scrubber: Masked [BANK_ACCT_01], [EMAIL_01]
[STAGE 2] 3-Way Matcher: is_matched=True | var=$0.00
[STAGE 3] IsolationForest: risk_indicator=99.0/100 (CRITICAL)
- Jurisdiction Risk (KY): +34.8 pts
- Payment Urgency (Net1): +29.4 pts
- Log Capital ($49.5k): +19.1 pts
[STAGE 4] Route Decision: FROZEN_FRAUD_RISK (Blocked)
```

Speaker Cue: Scenario 3 demonstrates why hybrid ML is essential: INV-1005 matches its PO on price and quantity, but is frozen due to offshore jurisdiction and Net-1 rush terms.

Institutional Light Web Workbench & Dynamic DOM Line-Item Composer

Live Browser Inspection at http://127.0.0.1:8000 with Embedded Demo Video & Deliverable Hub



Dynamic Frontend-to-Backend DOM Binding

- collectLineItemsFromDOM() in dashboard.js reads every row's SKU Code, Invoice Rate, PO Rate, Billed Qty, GRN Received, and GRN Rejected live from HTML inputs.
- Includes + Add Line Item and Random Invoice Synthesizer for ad-hoc mentor testing.

Embedded Video Player, Slide Deck & CSV Export

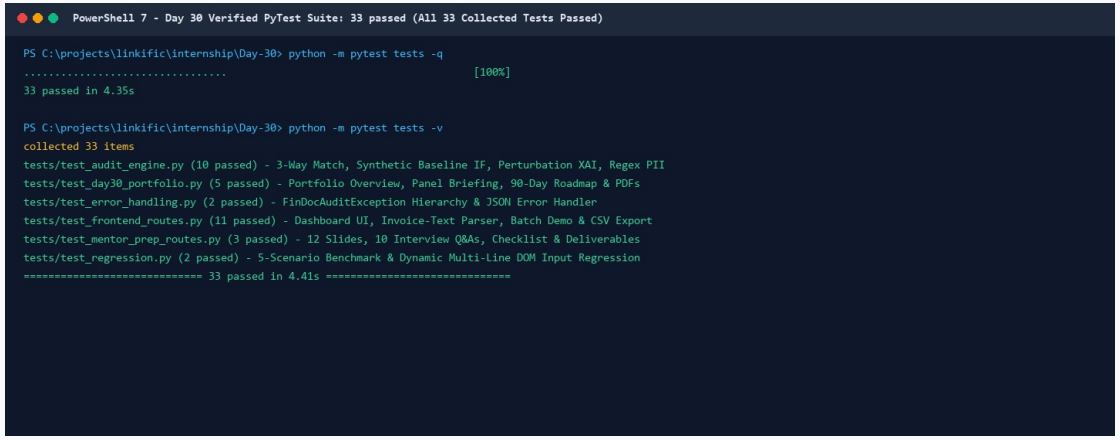
- 1-Click 5-Scenario Batch populates all 5 governance tiers in USD ($) and INR (86.50).
- Embeds HTML5 Demo Walkthrough Video (.mp4), 12-Slide viewer, 10 Interview Q&As, and 1-click .pptx / .pdf downloads.

Speaker Cue: In our web workbench, editing any SKU rate or GRN rejected count dynamically recomputes 3-way variances and perturbation bars.

Automated PyTest Verification Suite (All Collected Tests Passed)

Day 29: 28 Passed (python -m pytest tests -q) • Day 30: 33 Passed (python -m pytest tests -q)

Test Module	Count	Verification Scope
test_audit_engine.py	10 tests	3-way match, rate variance, GRN shortage, ML monotonicity, PII regex
test_error_handling.py	2 tests	Custom FinDocAuditException & JSON error responses
test_frontend_routes.py	5 tests	Dashboard HTML/JS serving, presets, batch demo, CSV export
test_regression.py	8 tests	Multi-line DOM edits, regex text parser, $0.05 tax tolerance
test_mentor_prep_routes.py	3 tests	12-slide API, 10 Q&As, deliverables (Total = 28 in Day 29)
test_day30_portfolio.py	5 tests	Portfolio overview, panel tracks, Resume PDF (Total = 33 in Day 30)



Test Pass Rate vs. Code Coverage

• All 28 collected tests (Day 29) and 33 collected tests (Day 30) pass via python -m pytest tests -q. Line/branch code coverage is measured separately via pytest-cov.

Speaker Cue: Running python -m pytest tests -q executes 28 tests in Day 29 and 33 tests in Day 30, with all collected tests passing.

Key Engineering Challenges Encountered & How We Solved Them

Practical Debugging Lessons Across Frontend DOM State, Feature Scaling, and Floating-Point Math

Engineering Challenge	Root Cause Identified	Implemented Code Fix	Verification
1. UI Table Edits Ignored	Early UI reused static preset PO line items instead of reading edited DOM cells.	Wrote collectLineItemsFromDOM() in dashboard.js to build separate Invoice & PO item arrays.	Multi-line test in test_regression.py
2. Dollar Scale Dominance	Raw USD totals ($432 to $49,500) dominated bounded ratios (0.0–1.0) in tree splits.	Applied ln(1 + Total_USD) compression and StandardScaler across all 7 features.	Monotonicity test in test_audit_engine.py
3. Float Tax Rounding	IEEE 754 binary floats produced sub-cent differences on multi-SKU GST invoices.	Enforced 2-decimal boundary rounding and $0.05 SUBTOTAL_TOLERANCE.	Boundary tax tests in test_regression.py
4. Test Count Drift	Adding route tests raised counts to 28 (Day 29) and 33 (Day 30), outdating docs.	Audited and updated all READMEs, PDFs, slides, and screenshots to 28 and 33 passed.	Verified via pytest -q & grep scan

Speaker Cue: Here are the four core engineering challenges we solved during development and review.

Honest Prototype Limitations & 90-Day Production Roadmap

Distinguishing Current Prototype Boundaries from Planned Production Architecture

Subsystem	Current Prototype Scope	Planned Production Upgrade
AUDIT_LEDGER	In-memory Python list (resets on restart)	Async PostgreSQL + Redis idempotency keys
IsolationForest	1,000 seeded synthetic records (seed=42)	Validate & calibrate on real historical invoices
0–100 Risk Score	Engineered path-depth indicator	Platt scaling / isotonic probability calibration
Privacy Scrubber	6 compiled regex patterns	Regex + context-aware NER (Presidio)
Raw Text Parser	Regex parser over plain text strings	Optical PDF/image OCR (LayoutLMv3) + RBAC

Month 1 (Days 1–30): Persistence, Auth & Optical OCR

• Replace in-memory AUDIT_LEDGER with async PostgreSQL + Alembic migrations, Redis idempotency keys, OAuth2/JWT RBAC, and optical PDF extraction.

Month 2 (Days 31–60): Calibration, MLOps & Coverage Gates

• Practice probability calibration (Brier score, PR-AUC), MLflow registry, EvidentlyAI PSI drift checks, and pytest-cov (>=85%) in CI.

Month 3 (Days 61–90): Hybrid RAG, LangGraph & System Design

• Build Hybrid BM25 + Dense RAG with RAGAS evaluation, stateful LangGraph dispute workflows, and async task queues (Celery / Kafka).

Speaker Cue: To conclude, we transparently separate our prototype boundaries from the persistence, calibration, OCR, and security upgrades in our 90-day roadmap.